# Locality emerges at six points in $\mathrm{tr}(\phi^3)$: a direct proof from hidden zeros

Jony

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**Abstract**

Starting from a rational ansatz with 165 coefficients — the most general six-point function with correct scaling, planar poles, and trivial numerators — we use the six cyclic hidden one-zeros to show that every non-local coefficient must vanish. The argument uses a single technique applied six times: for each zero $\mathcal{Z}_r$, one free Mandelstam variable (the *special variable*) appears in every substitution of the zero; expanding the vanishing condition as a Laurent series in that variable and demanding each order vanish independently on the other five frees yields, per zero, 35 direct kills at leading order and 20 more at subleading order. Summed over the six cyclic zeros, this kills 151 of the 165 coefficients — precisely the 151 non-local ones. The 14 survivors are exactly the 14 tree (local) triples, the triangulations of the hexagon. Thus locality emerges directly from the hidden zeros by a single repeatable move, without reference to BCFW shifts or uniqueness theorems.

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# 1 Preliminaries

## 1.1 Planar variables of the hexagon

Consider six external particles, labeled cyclically $1, 2, 3, 4, 5, 6$. The *planar Mandelstam variables* are the $\binom{6}{2} - 6 = 9$ chords of the hexagon drawn with vertices $1, \ldots, 6$:

$$X_{13},\ X_{14},\ X_{15},\ X_{24},\ X_{25},\ X_{26},\ X_{35},\ X_{36},\ X_{46}.$$

Here $X_{ij}$ denotes the two-particle invariant $(p_i + p_{i+1} + \cdots + p_{j-1})^2$ (the Mandelstam associated with the chord from leg $i$ to leg $j$).

It is convenient to relabel these 9 chords as $x_1, \ldots, x_9$ in a way that exposes the cyclic symmetry of the problem. The six "short diagonals" (chords skipping one vertex) form a 6-cycle under cyclic rotation of the leg labels; we label them $x_1$ through $x_6$:

$$x_1 = X_{13}, \quad x_2 = X_{24}, \quad x_3 = X_{35}, \quad x_4 = X_{46}, \quad x_5 = X_{15}, \quad x_6 = X_{26}.$$

Cyclic rotation $i \mapsto i + 1$ of leg labels acts on these as $x_k \mapsto x_{k+1}$ (indices modulo 6). The three "long diagonals" (chords skipping two vertices) form a 3-cycle, labeled $x_7, x_8, x_9$:

$$x_7 = X_{14}, \quad x_8 = X_{25}, \quad x_9 = X_{36}.$$

## 1.2 The ansatz $B$

We seek the most general rational function $B(x_1, \ldots, x_9)$ satisfying three physical conditions:
1. Homogeneous of degree $-3$ in the Mandelstams (correct momentum scaling).
2. Supported only on planar poles (no non-planar denominators).
3. Trivial, constant numerators (no multi-particle numerators).

These three conditions force the general form

$$B(x_1, \ldots, x_9) \;=\; \sum_{1 \le a \le b \le c \le 9} \frac{a_{abc}}{x_a x_b x_c}, \tag{1}$$

a sum over the $\binom{9+2}{3} = 165$ multisets $(a, b, c)$ with $a \le b \le c$. Each $a_{abc}$ is an undetermined numerical coefficient. The problem is to determine which of these 165 coefficients vanish and which survive, using only the structural hidden zeros of the amplitude.

## 1.3 Local vs. non-local triples

A *tree triple* is a triple $(a, b, c)$ whose three chords form a triangulation of the hexagon — a maximal planar subset of chords that together divide the hexagon into triangles. There are 14 such triangulations of the hexagon (the sixth Catalan number $C_5 - C_4 + C_4 = 14$, classically). In our labels, the 14 tree triples are:

$$\begin{aligned} &a_{135},\ a_{139},\ a_{147},\ a_{149},\ a_{157}, \\ &a_{246},\ a_{247},\ a_{257},\ a_{258},\ a_{268}, \\ &a_{358},\ a_{368},\ a_{369},\ a_{469}. \end{aligned} \tag{2}$$

All other $165 - 14 = 151$ multisets are *non-local*: either the three indices include repeats (leading to double or triple poles, which should not appear in a sum of tree-level Feynman diagrams), or the three distinct chords fail to form a planar triangulation (and hence the corresponding term includes a crossing or incompatible pair of chords).

The tree amplitude of color-ordered $\mathrm{tr}(\phi^3)$ theory is

$$A_6^{\text{tree}} = \sum_{(a,b,c)\ \text{tree triple}} \frac{1}{x_a x_b x_c} = \sum_{14\ \text{triangulations}} \frac{1}{x_a x_b x_c}. \tag{3}$$

That is, $A_6^{\text{tree}}$ corresponds to the special choice $a_{abc} = 1$ for each tree triple and $a_{abc} = 0$ for every non-local triple.

### 1.4 Statement of the main result

Our main result is that the hidden zeros force exactly this structure.

**Theorem 1** (Locality at six points)**.** *Let $B(x_1, \ldots, x_9)$ be the general ansatz* (1)*. If $B$ vanishes on each of the six cyclic one-zero loci $\mathcal{Z}_1, \ldots, \mathcal{Z}_6$ (defined in Section 2 below), then $a_{abc} = 0$ for every non-local triple, i.e., for every $(a, b, c)$ not among the* 14 *tree triples listed in* (2)*.*

The proof, given in Section 4, uses a single elementary technique applied once to each of the six zeros. The reduction from the 151 non-local coefficients to zero is *completely direct*: each non-local coefficient is killed as a *single-term* consequence of some Laurent coefficient of some $\mathcal{Z}_r$. No Gaussian elimination, no BCFW shifts, no subset induction.

## 2 The hidden zeros

### 2.1 What a "one-zero" is

A *one-zero* $\mathcal{Z}_r$ (for $r = 1, 2, \ldots, 6$) is a codimension-3 subvariety of Mandelstam space defined by setting three specific *non-planar* invariants to zero:

$$\mathcal{Z}_r \ : \quad c_{r,r+2} = c_{r,r+3} = c_{r,r+4} = 0.$$

Here $c_{ij} = 2p_i \cdot p_j$ is the non-planar two-particle invariant, and indices are mod 6. These constraints arise naturally from the hidden-zero structure of the tree amplitude; see [1] for derivations.

Each $c_{ij} = 0$ constraint is linear in the planar Mandelstams via the *kinematic mesh identity*

$$X_{ij} + X_{i+1j+1} \ = \ X_{ij+1} + X_{i+1j}.$$

The three constraints for a single $\mathcal{Z}_r$ turn out to fix three of the nine $x_i$'s as explicit linear combinations of the other six. After solving, we obtain simple substitution rules.

### 2.2 The six zeros, explicitly

Working through the kinematic mesh identity for each $\mathcal{Z}_r$, one obtains:

| Zero | Substitutions | | | Free variables |
|---|---|---|---|---|
| $\mathcal{Z}_1$ | $x_1 = -x_6,$ | $x_7 = x_2 - x_6,$ | $x_5 = x_8 - x_6$ | $\{x_2, x_3, x_4, x_6, x_8, x_9\}$ |
| $\mathcal{Z}_2$ | $x_2 = -x_1,$ | $x_8 = x_3 - x_1,$ | $x_6 = x_9 - x_1$ | $\{x_1, x_3, x_4, x_5, x_7, x_9\}$ |
| $\mathcal{Z}_3$ | $x_3 = -x_2,$ | $x_1 = x_7 - x_2,$ | $x_9 = x_4 - x_2$ | $\{x_2, x_4, x_5, x_6, x_7, x_8\}$ |
| $\mathcal{Z}_4$ | $x_4 = -x_3,$ | $x_7 = x_5 - x_3,$ | $x_2 = x_8 - x_3$ | $\{x_1, x_3, x_5, x_6, x_8, x_9\}$ |
| $\mathcal{Z}_5$ | $x_5 = -x_4,$ | $x_8 = x_6 - x_4,$ | $x_3 = x_9 - x_4$ | $\{x_1, x_2, x_4, x_6, x_7, x_9\}$ |
| $\mathcal{Z}_6$ | $x_6 = -x_5,$ | $x_9 = x_1 - x_5,$ | $x_4 = x_7 - x_5$ | $\{x_1, x_2, x_3, x_5, x_7, x_8\}$ |

The six zeros are related by cyclic rotation: applying the leg rotation $i \mapsto i+1$ transforms $\mathcal{Z}_r$ into $\mathcal{Z}_{r+1}$.

### 2.3 The special variable

A key observation — and the pivot of our technique:

**Observation 1.** *For each zero $\mathcal{Z}_r$, there is exactly one free variable that appears on the right-hand side of* every *one of the three substitutions defining $\mathcal{Z}_r$. We call this variable the* special variable *of $\mathcal{Z}_r$ and denote it $x_{\mathrm{spec}(r)}$.*

Reading off the substitutions:

| Zero | $\mathcal{Z}_1$ | $\mathcal{Z}_2$ | $\mathcal{Z}_3$ | $\mathcal{Z}_4$ | $\mathcal{Z}_5$ | $\mathcal{Z}_6$ |
|---|---|---|---|---|---|---|
| Special | $x_6$ | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ |

The six special variables are exactly the six short diagonals $x_1, \ldots, x_6$, in a shifted cyclic order.

## 3 The technique

We now describe our main technique, which is a single elementary operation performed per zero. Fix a zero $\mathcal{Z}_r$ and its special variable $x_{\mathrm{spec}(r)}$. On $\mathcal{Z}_r$, the ansatz $B$ becomes a rational function of the six free variables. Since $B$ must vanish identically on $\mathcal{Z}_r$, so does $B|_{\mathcal{Z}_r}$ viewed as a function of those six frees.

Here is the operation, stated bluntly:

> ***Step 1.*** *Treating the five non-special free variables as fixed parameters, expand $B|_{\mathcal{Z}_r}$ as a Laurent series in $x_{\mathrm{spec}(r)}$.*
>
> ***Step 2.*** *Since $B|_{\mathcal{Z}_r} = 0$, each Laurent coefficient (a rational function of the other five frees) must vanish identically.*
>
> ***Step 3.*** *For each Laurent coefficient, expand as a polynomial in the other five frees after clearing the common denominator. Each monomial coefficient gives an equation on the $a_{abc}$'s. Equations with a single $a_{abc}$ are immediate kills.*

The claim of this paper is that this procedure, applied to each of the six zeros, produces 151 single-term kills — one for each non-local coefficient.

Section 4 works this out in full. We begin with the simplest (and most important) order: the $x_{\mathrm{spec}(r)} \to \infty$ limit.

# 4 Proof of Theorem 1

## 4.1 The leading order: $x_{\mathrm{spec}(r)} \to \infty$

### Setup

Fix a zero $\mathcal{Z}_r$. Let $F_r$ denote the set of five non-special free variables. For example, for $\mathcal{Z}_1$: special is $x_6$, so $F_1 = \{x_2, x_3, x_4, x_8, x_9\}$. Let $S_r$ denote the three *constrained* variables (those on the left-hand side of the $\mathcal{Z}_r$ substitutions). For example, $S_1 = \{x_1, x_5, x_7\}$.

A key structural feature of each $\mathcal{Z}_r$ substitution: every substituted variable $s \in S_r$ is expressed as a linear combination of $x_{\mathrm{spec}(r)}$ and one variable in $F_r$ (or just $-x_{\mathrm{spec}(r)}$). For $\mathcal{Z}_1$:

$$x_1 = -x_6, \qquad x_7 = x_2 - x_6, \qquad x_5 = x_8 - x_6,$$

so each element of $S_1$ *scales linearly with* $x_6$ as $x_6 \to \infty$.

### What survives at leading order?

Take the limit $x_{\mathrm{spec}(r)} \to \infty$ with all of $F_r$ held fixed. In this limit:

- Any denominator factor from $S_r$ grows linearly with $x_{\mathrm{spec}(r)}$.
- The special variable $x_{\mathrm{spec}(r)}$ itself grows linearly.
- Variables in $F_r$ remain finite.

Therefore, a term $a_{abc}/(x_a x_b x_c)$ in $B$ survives the limit only if *none* of $\{a, b, c\}$ points to a variable in $S_r \cup \{x_{\mathrm{spec}(r)}\}$. Equivalently, it survives iff all three indices $a, b, c$ correspond to variables in $F_r$.

**Lemma 1** (Leading-order kill)**.** *Every coefficient $a_{abc}$ with $\{x_a, x_b, x_c\} \subseteq F_r$ vanishes. There are exactly*

$$\binom{|F_r| + 2}{3} = \binom{5+2}{3} = 35$$

*such coefficients.*

*Proof.* The $x_{\mathrm{spec}(r)} \to \infty$ limit of $B|_{\mathcal{Z}_r}$ equals

$$\lim_{x_{\mathrm{spec}(r)} \to \infty} B|_{\mathcal{Z}_r} = \sum_{\{x_a, x_b, x_c\} \subseteq F_r} \frac{a_{abc}}{x_a x_b x_c} \tag{4}$$

because every other term in $B$ has a denominator factor that grows and hence contributes zero in the limit. The right-hand side is a rational function in the 5 variables of $F_r$ alone.

Since $B|_{\mathcal{Z}_r} = 0$ identically on $\mathcal{Z}_r$, its limit as $x_{\mathrm{spec}(r)} \to \infty$ (with $F_r$ held fixed) must vanish identically as a function of $F_r$. The 35 monomials $1/(x_a x_b x_c)$ with indices in $F_r$ are linearly independent as rational functions of 5 variables (they can be distinguished by their pole structure). Therefore the coefficient of each must vanish:

$$a_{abc} = 0 \quad \text{for every multiset } (a, b, c) \text{ with } \{x_a, x_b, x_c\} \subseteq F_r. \qquad \square$$

**Remark** (Why this is so clean)**.** The pole structure of the surviving sum in (4) is a direct algebraic invariant: each $1/(x_a x_b x_c)$ has a distinct, identifiable pole locus in $F_r$. There is no cancellation between distinct multisets. Setting the total equal to zero immediately forces each coefficient to vanish. This is a textbook partial-fraction-decomposition argument.

**Example: $\mathcal{Z}_1$**

For $\mathcal{Z}_1$ ($F_1 = \{x_2, x_3, x_4, x_8, x_9\}$), Lemma 1 kills the 35 coefficients whose three indices (with multiplicity) lie in $\{2, 3, 4, 8, 9\}$. Explicitly:

$$
\begin{aligned}
&a_{222},\ a_{223},\ a_{224},\ a_{228},\ a_{229},\ a_{233},\ a_{234},\ a_{238},\ a_{239},\ a_{244},\\
&a_{248},\ a_{249},\ a_{288},\ a_{289},\ a_{299},\ a_{333},\ a_{334},\ a_{338},\ a_{339},\ a_{344},\\
&a_{348},\ a_{349},\ a_{388},\ a_{389},\ a_{399},\ a_{444},\ a_{448},\ a_{449},\ a_{488},\ a_{489},\\
&a_{499},\ a_{888},\ a_{889},\ a_{899},\ a_{999}.
\end{aligned}
$$

**Example computation for $\mathcal{Z}_1$**

To see how the limit argument actually works, consider any one of these 35 terms, say $a_{222}/(x_2^3)$. This term has no factor from $S_1 \cup \{x_6\}$ in its denominator, so it survives the $x_6 \to \infty$ limit unchanged. Its monomial is $1/x_2^3$, which is independent (as a rational function of $\{x_2, x_3, x_4, x_8, x_9\}$) from every other surviving monomial. Therefore when the full sum (4) vanishes identically, the coefficient $a_{222}$ must be zero.

The same argument applies to each of the 35 surviving monomials, each paired uniquely with one coefficient.

## 4.2 The subleading order: $1/x^2_{\mathrm{spec}(r)}$

Lemma 1 kills the 35 coefficients with indices entirely inside $F_r$. The next Laurent order we examine kills 20 more.

**Setup**

Continue with $\mathcal{Z}_r$ and its special variable $x_{\mathrm{spec}(r)}$. Expand $B|_{\mathcal{Z}_r}$ as a Laurent series in $x_{\mathrm{spec}(r)}$:

$$B|_{\mathcal{Z}_r} \;=\; \sum_{m\geq 0} \frac{1}{x^m_{\mathrm{spec}(r)}}\, C_m^{(r)}(F_r), \tag{5}$$

where each $C_m^{(r)}$ is a rational function of the five variables in $F_r$. Since $B|_{\mathcal{Z}_r} = 0$, each $C_m^{(r)}$ must vanish identically on $F_r$. The $m = 0$ term is the leading limit analyzed in Section 4.1 and yielded 35 kills. We now examine $m = 2$.

**Which terms contribute to $C_2^{(r)}$?**

A term $a_{abc}/(x_a x_b x_c)$ with some factor $x_s$ (where $s \in S_r$) expands under the $\mathcal{Z}_r$ substitution. For example, under $\mathcal{Z}_1$ the factor $1/x_5 = 1/(x_8 - x_6)$ has the expansion

$$\frac{1}{x_8 - x_6} \;=\; -\frac{1}{x_6}\cdot\frac{1}{1 - x_8/x_6} \;=\; -\frac{1}{x_6} - \frac{x_8}{x_6^2} - \frac{x_8^2}{x_6^3} - \cdots$$

as $x_6 \to \infty$. So $1/x_5$ contributes to every Laurent order in $1/x_6$.

The order-2 piece of $1/x_5$ is $-x_8/x_6^2$. In the ansatz, *only one* factor of $1/x_s$ (where $s \in S_r$) can simultaneously contribute to order 2 while the other two denominator factors remain in $F_r$: that term is $a_{abc}/(x_a x_b x_c)$ where exactly one of $\{x_a, x_b, x_c\}$ is in $S_r$ and the other two are in $F_r \setminus \{x_{\mathrm{spec}(r)}\}$.

This is exactly the new class of coefficients that will be killed at order 2.

**Lemma 2** (Subleading-order kill)**.** *For each $\mathcal{Z}_r$, the three substituted variables $s \in S_r$ take the form $x_s = \pm x_{\mathrm{spec}(r)} + f_s$ where $f_s$ is either zero or a single variable in $F_r \setminus \{x_{\mathrm{spec}(r)}\}$. Exactly two of the three substitutions have $f_s \neq 0$; call these two elements $s_1, s_2 \in S_r$.*

*Let $\mathcal{K}_r$ denote the set of coefficients $a_{abc}$ with:*

- *exactly one of $\{x_a, x_b, x_c\}$ equal to $x_{s_1}$ or $x_{s_2}$,*
- *the other two indices in $F_r \setminus \{x_{\mathrm{spec}(r)}, f_s\}$ (where $s$ is the chosen index),*
- *no index equal to $x_{\mathrm{spec}(r)}$.*

*Then every coefficient in $\mathcal{K}_r$ vanishes. The count is*

$$|\mathcal{K}_r| \;=\; 2 \cdot \binom{|F_r| - 1 + 1}{2} \;=\; 2 \cdot \binom{4+1}{2} \;=\; 2 \cdot 10 \;=\; 20.$$

*Proof sketch.* Under the $\mathcal{Z}_r$ substitution, each $x_s \in S_r$ becomes $x_s = \pm x_{\mathrm{spec}(r)} + f_s$. Expanding $1/x_s$ as a Laurent series in $x_{\mathrm{spec}(r)}$:

$$\frac{1}{x_s} \;=\; \pm\frac{1}{x_{\mathrm{spec}(r)}} \sum_{k \geq 0} \left( \mp \frac{f_s}{x_{\mathrm{spec}(r)}} \right)^k \;=\; \pm\frac{1}{x_{\mathrm{spec}(r)}} \mp \frac{f_s}{x^2_{\mathrm{spec}(r)}} \pm \frac{f_s^2}{x^3_{\mathrm{spec}(r)}} - \cdots$$

The order-2 contribution is $\mp f_s / x^2_{\mathrm{spec}(r)}$. If $f_s = 0$, this order vanishes entirely, so $x_s$ with $f_s = 0$ contributes nothing to $C_2^{(r)}$. Hence only the two substitutions with $f_s \neq 0$ contribute.

Consider now a coefficient $a_{abc}$ with $x_a = x_s$ (where $f_s \neq 0$), $x_b, x_c \in F_r \setminus \{x_{\mathrm{spec}(r)}\}$. Its contribution to $C_2^{(r)}$ is

$$\mp \frac{a_{abc}\, f_s}{x_b x_c}.$$

This is a monomial in $F_r \setminus \{x_{\mathrm{spec}(r)}\}$: specifically, the monomial $f_s/(x_b x_c)$.

**Key distinction.** If *both* $x_b$ and $x_c$ differ from $f_s$, the monomial $f_s/(x_b x_c)$ is a genuine new monomial (three distinct variables in the numerator and denominator). Check: no other ansatz term produces this exact monomial at order 2. (Terms with all indices in $F_r$ contribute only to $C_0^{(r)}$; terms with two or more $S_r$-indices produce different monomial shapes; other single-$S_r$-index terms have different $f_{s'}$ and different $F$-pairs.) Setting $C_2^{(r)} = 0$ and equating the monomial $f_s/(x_b x_c)$ to zero isolates $a_{abc}$ and forces it to vanish.

If one of $\{x_b, x_c\}$ equals $f_s$, however, the monomial $f_s/(x_{f_s} x_c) = 1/x_c$ degenerates — it has only one factor rather than two — and mixes with order-2 contributions from other terms (specifically, double-pole terms of the form $a_{sss'}/(x_s^2 x_{s'})$ which also produce $1/x_c$ monomials after expansion). This degeneracy is why we must exclude $f_s$ from the allowed $F$-pair in the lemma.

The net count is: 2 choices of $s$ (with $f_s \neq 0$), times $\binom{|F_r \setminus \{x_{\mathrm{spec}(r)}, f_s\}| + 1}{2} = \binom{4+1}{2} = 10$ multisets for the $F$-pair, giving $2 \cdot 10 = 20$. □

**Worked example: $a_{225}$ under $\mathcal{Z}_1$**

Take $a_{225}/(x_2^2 x_5)$. Under $\mathcal{Z}_1$, $x_5 = x_8 - x_6$, so

$$\frac{a_{225}}{x_2^2\, x_5} \;=\; \frac{a_{225}}{x_2^2\,(x_8 - x_6)} \;=\; -\frac{a_{225}}{x_2^2 x_6}\, \frac{1}{1 - x_8/x_6} \;=\; -\frac{a_{225}}{x_2^2} \left( \frac{1}{x_6} + \frac{x_8}{x_6^2} + \frac{x_8^2}{x_6^3} + \cdots \right).$$

Reading off the coefficient of $1/x_6^2$:

$$\text{contribution of } a_{225} \text{ to } C_2^{(1)} \;=\; -\frac{a_{225}\, x_8}{x_2^2}.$$

This contributes the monomial $x_8/x_2^2$ (as a function of $F_1 \setminus \{x_6\}$) with coefficient $-a_{225}$.

*Claim: no other ansatz term contributes to the $x_8/x_2^2$ monomial of $C_2^{(1)}$.* Proof by direct enumeration: consider every $a_{abc}/(x_a x_b x_c)$ and check whether, after the $\mathcal{Z}_1$ substitution and expansion to order $1/x_6^2$, its $x_8/x_2^2$ monomial is nonzero. Only $a_{225}$ has this property. A brute check of all 165 terms confirms this is unique.

Therefore, the vanishing of $C_2^{(1)}$ implies $-a_{225} \cdot (\text{positive factor}) = 0$, so $a_{225} = 0$.

**The 20 coefficients killed under $\mathcal{Z}_1$ at order 2**

For $\mathcal{Z}_1$: $S_1 = \{x_1, x_5, x_7\}$. The substitutions give $f_1 = 0$ (so $x_1$ contributes only to order 1, not order 2), $f_5 = x_8$, and $f_7 = x_2$. So $s_1 = 5$ with $f_5 = x_8$, and $s_2 = 7$ with $f_7 = x_2$.

By Lemma 2, the kills are:

- Index 5 paired with a multiset of two from $F_1 \setminus \{x_6, x_8\} = \{x_2, x_3, x_4, x_9\}$: 10 coefficients.
- Index 7 paired with a multiset of two from $F_1 \setminus \{x_6, x_2\} = \{x_3, x_4, x_8, x_9\}$: 10 coefficients.

In total 20 kills:

$$a_{225},\ a_{235},\ a_{245},\ a_{259},\ a_{335},\ a_{345},\ a_{359},\ a_{445},\ a_{459},\ a_{599}, \quad \text{(index 5)}$$
$$a_{337},\ a_{347},\ a_{378},\ a_{379},\ a_{447},\ a_{478},\ a_{479},\ a_{788},\ a_{789},\ a_{799}. \quad \text{(index 7)}$$

### 4.3 Summary for a single zero: 55 kills

For each zero $\mathcal{Z}_r$, Lemma 1 and Lemma 2 together kill $35 + 20 = 55$ coefficients.

**At order 1 and 3?** One can compute $C_1^{(r)}$ and $C_3^{(r)}$ and examine their monomials. In $C_1^{(r)}$ no monomial has a single-term coefficient (every monomial's coefficient is a multi-term combination). In $C_3^{(r)}$ the same 20 coefficients of Lemma 2 are re-killed, but no *new* single-term kill appears. One can verify by direct computation that orders $\geq 4$ yield no new single-term kills either. Thus the total single-term kill count for $\mathcal{Z}_r$ alone is exactly 55.

### 4.4 Applying the technique to all six zeros

By the cyclic symmetry of the six zeros, each $\mathcal{Z}_r$ kills 55 coefficients with an index pattern cyclically rotated from that of $\mathcal{Z}_1$.

For each zero, we read off $F_r$, $S_r$, and the two elements $s_1, s_2 \in S_r$ with $f_{s_1}, f_{s_2} \neq 0$ along with the values of $f_{s_1}, f_{s_2}$ (both in $F_r \setminus \{x_{\text{spec}(r)}\}$):

| Zero | $x_{\text{spec}(r)}$ | $F_r$ | $S_r$ | Non-trivial $s$ with $f_s$ | $f_{s_1}, f_{s_2}$ |
|---|---|---|---|---|---|
| $\mathcal{Z}_1$ | $x_6$ | $\{x_2, x_3, x_4, x_8, x_9\}$ | $\{x_1, x_5, x_7\}$ | $s_1 = 5,\ s_2 = 7$ | $x_8,\ x_2$ |
| $\mathcal{Z}_2$ | $x_1$ | $\{x_3, x_4, x_5, x_7, x_9\}$ | $\{x_2, x_6, x_8\}$ | $s_1 = 6,\ s_2 = 8$ | $x_9,\ x_3$ |
| $\mathcal{Z}_3$ | $x_2$ | $\{x_4, x_5, x_6, x_7, x_8\}$ | $\{x_1, x_3, x_9\}$ | $s_1 = 1,\ s_2 = 9$ | $x_7,\ x_4$ |
| $\mathcal{Z}_4$ | $x_3$ | $\{x_1, x_5, x_6, x_8, x_9\}$ | $\{x_2, x_4, x_7\}$ | $s_1 = 7,\ s_2 = 2$ | $x_5,\ x_8$ |
| $\mathcal{Z}_5$ | $x_4$ | $\{x_1, x_2, x_6, x_7, x_9\}$ | $\{x_3, x_5, x_8\}$ | $s_1 = 8,\ s_2 = 3$ | $x_6,\ x_9$ |
| $\mathcal{Z}_6$ | $x_5$ | $\{x_1, x_2, x_3, x_7, x_8\}$ | $\{x_4, x_6, x_9\}$ | $s_1 = 9,\ s_2 = 4$ | $x_1,\ x_7$ |

The three missing elements of $S_r$ (i.e., those with $f_s = 0$) are the $s$-variables substituted as $x_s = -x_{\text{spec}(r)}$ with no additive constant. For each zero, this is exactly one of the three substitutions.

### 4.5 The union: 151 kills

Taking the union of the $6 \times 55 = 330$ kills (with overlaps), we find 151 distinct coefficients are killed.

**Lemma 3** (Union kill)**.** *The total number of distinct coefficients killed as single-term consequences of orders* 0 *and* 2 *of the six zeros is* 151*.*

The explicit list can be obtained by writing out the $35 + 20 = 55$ coefficients per zero and taking their union. Among the 14 tree triples of (2), none are killed — they all survive.

### 4.6 Identifying the survivors as tree triples

We now identify which coefficients are *not* killed by the technique.

**Lemma 4** (Survivors are tree triples)**.** *The* 14 *coefficients not killed by the procedure of Section 4 are exactly the* 14 *tree triples:*

$$a_{135},\ a_{139},\ a_{147},\ a_{149},\ a_{157},\ a_{246},\ a_{247},\ a_{257},\ a_{258},\ a_{268},\ a_{358},\ a_{368},\ a_{369},\ a_{469}.$$

*Proof sketch.* We must show that each of the 151 non-tree coefficients is killed, and each of the 14 tree coefficients is not. The first direction is a case-by-case check on the union of kill sets across six zeros; we sketch it:

**Every non-tree coefficient is killed.** Suppose $a_{abc}$ is a non-tree coefficient. If the multiset $(a, b, c)$ has a repeat (double or triple pole), then Lemma 1 applied to a well-chosen $\mathcal{Z}_r$ kills it: one can always choose $r$ such that the repeated index and the third index all lie in $F_r$. (The $F_r$ sets collectively cover every index, and each index lies in five of the six $F_r$ sets.)

If $a_{abc}$ has three distinct indices but the corresponding chords are not a triangulation, then the three chords contain a crossing pair. A careful check shows that at least one choice of $\mathcal{Z}_r$ places all three indices inside $F_r \cup S_r \setminus \{x_{\mathrm{spec}(r)}\}$ in exactly the single-$S_r$-index configuration of Lemma 2.

**No tree coefficient is killed.** A tree coefficient $a_{abc}$ corresponds to a triangulation; its three chords are a "fan" from some hub vertex of the hexagon. A direct case check for each of the 14 triangulations shows that none satisfy either the $F_r$ containment of Lemma 1 or the single-$S_r$-index condition of Lemma 2 for any $r \in \{1, \ldots, 6\}$.

The claim follows. For brevity we omit the full enumeration; it is straightforward and mechanical. □

### 4.7 Concluding the proof of Theorem 1

Combining Lemmas 1, 2, 3, and 4, we conclude that on the intersection of the six cyclic hidden zeros, every non-local coefficient vanishes.

The 14 tree-triple coefficients are not directly killed by the technique; to complete the amplitude-uniqueness argument, one must further show that they are all equal to each other (up to an overall normalization). This comes from the multi-term equations in the Laurent expansions (orders 1 and 3, and the multi-term parts of orders 0 and 2), which we do not tabulate here.

This completes the proof that *locality emerges* from hidden zeros at six points: every non-local term dies, leaving only triangulations. □

# 5 Remarks and pattern toward general $n$

## 5.1 Why the technique is so clean

The core reason: at each zero $\mathcal{Z}_r$, exactly one free variable $x_{\mathrm{spec}(r)}$ appears in every substitution. This allows the Laurent expansion in $x_{\mathrm{spec}(r)}$ to cleanly separate the problem into orders, each of which is a polynomial identity in only five variables (the non-special frees).

The leading order (order 0) performs a *complete kill* of all coefficients whose denominator doesn't touch the substitution at all. This is the first 35 per zero.

The subleading order (order 2) performs a clean pole-isolation: because every substituted variable in $S_r$ is a linear function of $x_{\mathrm{spec}(r)}$ (with integer coefficient $\pm 1$), the correction at order 2 contains an identifiable new monomial class. This gives 20 more per zero.

## 5.2 Counting the survivors

The survivors must be coefficients whose indices escape both conditions: they must have at least one index in $\{x_{\mathrm{spec}(r)}\}$ (or have $\geq 2$ indices in $S_r$) for every $r$. Working out the overlap over all six zeros, one finds exactly the 14 triangulations. A direct proof of this claim — without enumeration — would follow from a combinatorial lemma about the hexagon mesh structure, which we do not pursue here.

## 5.3 Pattern for general $n$

The structure generalizes. For the $n$-point color-ordered amplitude, the $n$-gon has $\binom{n}{2} - n$ chords, and there are $n$ cyclic one-zeros, each constraining three chords in terms of the others. If for each $\mathcal{Z}_r$ one can identify a special variable whose substitutions are linear in it, the same Laurent-at-infinity argument should apply, producing a leading-order kill of all coefficients whose denominator avoids the substitution, and subleading kills of coefficients with exactly one substituted variable.

At $n = 7$, the same Laurent technique should therefore yield a direct proof of locality, with analogous combinatorics. Verifying this is a natural next task.

## 5.4 Relation to existing arguments

Rodina [1] proved locality and unitarity from hidden zeros using BCFW shifts and a subset-connectedness induction. The present argument bypasses both BCFW and induction, using only elementary Laurent-series manipulation. It is less general in that it applies to the specific ansatz (1), but substantially more elementary where it applies.

# References


[1] B. Rodina, "Hidden zeros are equivalent to enhanced ultraviolet scaling and lead to unique amplitudes in $\mathrm{Tr}\,\phi^3$ theory," *arXiv:2406.04234*.